

Affine Gap Alignment

Inputs: $\langle X, Y, \alpha, \mu, \gamma, \tau \rangle$

Output: α (# match)

- β (# mismatch)

- γ (# clusters)

- τ (# single indels)

Matrices: V, G, E, F

$V(0,0) \leftarrow 0$

for $i=1$ to m : $V(i,0) \leftarrow -\gamma - i\tau$

for $j=1$ to n : $V(0,j) \leftarrow -\gamma - j\tau$

for $i=1$ to m , $j=1$ to n :

$$G(i,j) \leftarrow \begin{cases} V(i-1,j-1) + \alpha, & \text{if } x_i = y_j \\ V(i-1,j-1) - \mu, & \text{if } x_i \neq y_j \end{cases}$$

$$E(i,j) = \max \begin{cases} E(i,j-1) - \tau \\ V(i,j-1) - \gamma - \tau \end{cases}$$

$$F(i,j) = \max \begin{cases} F(i-1,j) - \tau \\ V(i-1,j) - \gamma - \tau \end{cases}$$

$$V(i,j) = \max \{ G(i,j), E(i,j), F(i,j) \}$$

X, Y : strings of length m, n , indexed by i, j

α : match score

μ : mismatch penalty

γ : gap opening penalty

τ : gap extension penalty

V : best-score matrix

G : match | mismatch matrix

E : X-gap matrix

F : Y-gap matrix

time complexity: $O(mn)$ quadratic

traceback:

use back pointers in V to make moves

• if it was $G \rightarrow$ move up, left

• if it was $F \rightarrow$ move left (Y gap)

• if it was $E \rightarrow$ move up (X gap)

Statistical Basis of Local Alignment

Length of optimal global of 2 random seq: $O(N)$

Length of optimal local of 2 random seq: $O(\log N)$

$\hookrightarrow$ stopping at length $O(\log N)$ is optimal because of special scoring scheme

Karlin-Altschul Statistics for ungapped alignment (5 axioms)

① scoring matrix needs at least 1 positive score

② the expected score of the scoring matrix must be negative

③ the letters of the random sequences are independent and identically distributed

④ sequences are infinitely long

⑤ alignments do not contain gaps

Maximum local alignment such that extending segment decreases total score

Karlin-Altschul Equation: $E = kmne^{-\lambda S}$

Database search of size n with query of length m

E = # alignments expected by chance for alignments with score S

Dayhoff log odds ratio score: $\text{LOD}(\text{pair of amino acids}) = \log_2 \left(\frac{\text{observed freq}}{\text{expected freq}} \right)$

$S_{i,j} = \frac{q_{i,j}}{p_i p_j}$ where p_i, p_j are freq of i and j and $q_{i,j}$ is distribution of aa in alignment

if observed freq = expected freq, $LOD(i,j) = 0$

if $LOD > 0$, pairing i and j in alignment is common

if $LOD < 0$, pairing i and j in alignment is unlikely